MAGGIE WU

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EDUCATION

CARNEGIE MELLON UNIVERSITY | Pittsburgh, PA *Anticipated Graduation: May 2028*
Candidate for Bachelor of Science in Electrical & Computer Engineering GPA: 3.84/4.00
Additional Major in Robotics & Minor in Computer Science

EXPERIENCE

SURGICAL MECHATRONICS LAB **Carnegie Mellon University, PA**
Researcher *May 2025 – Present*

- Developed the physical assembly of the HeartPrinter, a parallel wire robot for epicardial interventions

ENVIRONMENTAL ENGINEERING AND SCIENCE LAB **Stony Brook University, NY**
Research Intern *July 2023 – November 2023*

- Characterized denitrification kinetics and greenhouse gas emission of woodchip bioreactors used in wastewater treatment
- Co-authored publication in Water Research: https://doi.org/10.1016/j.watres.2024.122562
- Named Regeneron STS Top 300 Scholar; Won Third Place at New York State Science & Engineering Fair

TONGE LAB **Stony Brook University, NY**
Research Intern *July 2022 – August 2022*

- Worked to synthesize a PROTAC to selectively degrade the main protease of SARS-CoV-2
- Prepared and worked up chemical reactions; learned mass spectrometry and NMR spectroscopy

PROJECT EXPERIENCE

HEARTPRINTER | CARNEGIE MELLON UNIVERSITY *May 2025 – Present*

- Designed and assembled the power distribution and control unit, housed in a portable 4U rack case
- Constructed the daughter unit (consisting of motors, load cells, etc), which resides next to the patient's incision

THE WONDER GLOVE | CARNEGIE MELLON UNIVERSITY *March 2025*

- Designed and constructed a hockey glove employing a flexible wrist guard to minimize wrist injuries without inhibiting movement, along with a replaceable palm to improve durability and affordability
- Named MVP of a team of 4 in Rethink the Rink make-a-thon sponsored by Covestro and the Pittsburgh Penguins

AFFORDABLE STREET NAVIGATION DEVICE FOR VISUALLY-IMPAIRED PEDESTRIANS *February 2023 – June 2023*

- Adapted physical prototype (below) for street navigation by creating a real-time sidewalk, crosswalk, and vehicle localization model trained on a custom dataset, reducing costs from commercial alternatives by over 90%
- Named National Finalist in JYEM Research Paper Competition (Top 5%); Presented at JYEM Expo

FACIAL RECOGNITION HAT FOR LOW-VISION USERS *December 2022 – February 2023*

- Collaborated with a team of 3 and a low-vision co-designer to construct a hat integrating a camera, earbuds, and Raspberry Pi computer to identify and verbally communicate the names of people in view
- Awarded $125 funding in the MIT BeaverWorks CRE[AT]E Assistive Technology Challenge

LEADERSHIP EXPERIENCE, AWARDS, SKILLS

NCAA DIVISION 3 TRACK & FIELD ATHLETE *August 2024 – Present*

- Attended 2-hour daily practices and competed at weekly meets (throughout PA, WV, OH, NY) in hurdle and jump events

GREAT NECK SOUTH HIGH SCHOOL SCIENCE OLYMPIAD *October 2021 – June 2024*

- As Treasurer, organized 10+ bake sales, raising $1000+
- Won 18 Medals (Regional, State, National), including: 1st Place Wi-Fi Lab (constructed Double Biquad antenna), 1st Place Detector (made accurate force scale with self-made load cell), 2nd Place Trajectory (built accurate catapult for ping-pong balls)

SELECTED AWARDS: National Merit $2500 Scholarship Recipient, Regeneron STS Top 300 Scholar

SKILLS: Programming: Python, C, C++, JavaScript, React, Next.js, CSS, x86-64 Assembly, Linux, Arduino, Raspberry Pi, Visual Studio, MATLAB (enrolled for Fall 2025) | CAD: Fusion 360, SolidWorks, Inventor | Fabrication: Soldering, Laser Cutting, 3D Printing | Data analysis: ANOVA, Excel